

ELECTROPRO

QUANTUM & DEEPTech EDUCATION PROGRAM

2026-27 SCHOOL PARTNERSHIP PROPOSAL

FROM BITS TO QUBITS

Preparing students for the next generation of technology

Technology & Innovation Education Initiative by Electropo India

Designed for CBSE schools, school managements, principals, innovation labs and future-skills programs.

1. Executive Summary

Electropro India proposes a structured 2026–27 Quantum & DeepTech Education Program for schools. The program introduces students to quantum computing through age-appropriate concepts, visual demonstrations, simulations, coding and project-based learning.

Program promise

Learn • Experiment • Code • Innovate

2. About Electropro India

Electropro India is positioned as a technology education and innovation partner for schools, with a focus on emerging technologies and hands-on learning.

Core capability areas

Robotics • Electronics • Embedded Systems • IoT • Artificial Intelligence • Drones • Coding • Innovation Projects

3. Why Quantum & DeepTech Education?

Quantum technologies are moving from research laboratories toward practical experimentation, education and industry development. Students benefit from early conceptual exposure without needing advanced physics prerequisites.

For schools

Future-skills positioning, STEM enrichment, innovation culture, project-based learning and differentiated technology offerings.

For students

Understanding the evolution of computing, computational thinking, quantum circuits and emerging technology careers.

4. Vision & Mission

Vision

Build the next generation of Quantum and DeepTech innovators from the classroom.

Mission

Make emerging technology understandable, practical and inspiring through structured curriculum, demonstrations, coding, projects and mentorship.

Principles

Age appropriate • Hands-on • Project based • Industry aware • Future focused • Accessible

5. Program Architecture

Five levels cover student and teacher development from awareness to advanced programming.

Level	Classes	Sessions	Focus
Quantum Explorer	6-7	20	Concepts + activities
Quantum Innovator	8-9	25	Gates + circuits + projects
Foundation	10	30	Algorithms + security + AI
Quantum Programmer	11-12	36	Python + programming + capstone
Quantum Educator	Teachers	12	Delivery + practicals + assessment

6. Student Learning Pathway

Level 1 — Quantum Explorer

Classes 6-7 | 20 sessions | Concepts, demonstrations, qubits, superposition, entanglement and applications.

Level 2 — Quantum Innovator

Classes 8-9 | 25 sessions | Quantum gates, circuits, security, simulations and mini projects.

Level 3 — Quantum Computing Foundation

Class 10 | 30 sessions | Fundamentals, algorithms, cybersecurity and Quantum + AI.

Level 4 — Quantum Programmer

Classes 11-12 | 36 sessions | Python, quantum programming, algorithms, applications and capstone.

Level 5 — Quantum Educator

Teachers | 12 sessions | Concepts, activities, classroom delivery, practicals and assessment.

7. Level 1 — Quantum Explorer

Module 1

Quantum science; atoms and particles; classical vs quantum world; quantum technology.

Module 2

Bits, binary and qubits; interactive bit-vs-qubit activities.

Module 3

Superposition, measurement, entanglement and quantum randomness.

Module 4

Quantum computers, sensors, communication, security and careers.

Capstone

Build a physical or digital Qubit Model.

8. Level 2 — Quantum Innovator

Foundation

Quantum computing, qubits, superposition, measurement and entanglement.

Quantum gates

X, H, Z and CNOT gates; circuit diagrams and combinations.

Applications

Quantum cryptography, communication, sensors, simulation and Quantum + AI.

Hands-on

First circuit, quantum coin toss, entanglement simulation and random number generator.

9. Level 3 — Class 10 Foundation

Unit 1

Quantum fundamentals, computing history, qubits, Bloch Sphere introduction, superposition, measurement and entanglement.

Unit 2

X, Y, Z, Hadamard, CNOT and controlled gates; circuit construction.

Unit 3

Deutsch, Grover, teleportation and Shor concepts.

Unit 4

Classical cryptography, quantum key distribution and future cybersecurity.

Unit 5

Quantum + AI, optimization and machine learning concepts.

Capstone

Design a Quantum Solution for a Real-World Problem.

10. Level 4 — Quantum Programmer

Python foundation

Variables, conditions, loops, functions, lists, dictionaries and mathematical operations.

Quantum programming

Qubits, circuits, gates, measurement, visualization and simulator-based execution.

Practical sequence

Quantum coin toss; random number generation; Bell state; entanglement; teleportation.

Algorithms

Deutsch-Jozsa, Grover, Quantum Fourier Transform introduction and Shor concept.

Capstone

Team-based Quantum Innovation Project.

11. Ten Core Quantum Practicals

1

Classical Bit Simulator

2

Qubit Simulator

3

Quantum Coin Toss

4

Superposition Demonstration

5

Hadamard Gate Circuit

6

Bell State / Entanglement

7

Quantum Random Number Generator

8

Quantum Teleportation Simulation

9

Grover Search Demonstration

10

Quantum Cryptography Demonstration

No.	Practical
1	Classical Bit Simulator
2	Qubit Simulator
3	Quantum Coin Toss
4	Superposition Demonstration
5	Hadamard Gate Circuit
6	Bell State / Entanglement
7	Quantum Random Number Generator
8	Quantum Teleportation Simulation
9	Grover Search Demonstration
10	Quantum Cryptography Demonstration

12. Electropro Quantum Lab

Starter lab

Computer/laptop, internet connection, Python environment, quantum simulator, project notebook and presentation tools.

Advanced lab

Cloud-accessible quantum computing, quantum simulators, AI workstation, electronics/IoT equipment, robotics equipment and demonstration kits.

Implementation

Begin with software-based learning and upgrade progressively.

13. Electropro Quantum Club

Purpose

Create a sustained school community around quantum computing and emerging technologies.

Monthly

Quantum Talk • Quantum Challenge • Coding Activity

Quarterly

Project Exhibition • Quiz • Mini Hackathon

Annual

Electropro Quantum Day with exhibitions, coding, demonstrations and career sessions.

14. Teacher Development

Program

Electropro Certified Quantum Educator — a 12-session program covering fundamentals, demonstrations, gates, circuits, programming environments, projects and assessment.

Outcome

Teachers gain practical confidence to facilitate age-appropriate activities and coordinate the Quantum Club.

Certificate

Electropro Certified Quantum Educator — CQE 2026-27.

15. Certification & Assessment

Student certificates

Quantum Explorer (6–7); Quantum Innovator (8–10); Quantum Programmer (11–12).

Assessment

Concept Knowledge 20; Practical Work 25; Coding 20; Project 25; Presentation 10. Total 100.

16. Innovation Project Framework

Stage 1

Identify a problem or opportunity.

Stage 2

Research the classical approach.

Stage 3

Ask where quantum methods could potentially contribute.

Stage 4

Build a simulation or conceptual prototype.

Stage 5

Document assumptions, results and limitations.

Stage 6

Present to teachers, mentors and peers.

17. School Implementation Model

Phase 1

Management/principal briefing, teacher orientation and student awareness.

Phase 2

Timetable integration, baseline assessment and student onboarding.

Phase 3

Weekly classes, practical sessions, projects and club activities.

Phase 4

Project exhibition, assessment and certification.

Phase 5

Teacher development, advanced lab and inter-school competitions.

18. Partnership Models

Model A — Curriculum Partner

Curriculum, trainer support, practical content and assessments.

Model B — Lab & Program Partner

Curriculum plus lab setup, demonstrations, clubs and teacher training.

Model C — Centre of Excellence

Joint advanced Quantum & DeepTech learning centre.

Commercial approach

Fees can be customized by student count, grades, trainer visits, lab requirements, certification and project scope.

19. 2026-27 Roadmap

Q1

Curriculum finalization • Teacher orientation • School launch • Awareness sessions.

Q2

Regular instruction • First practicals • Quantum Club • Student mini projects.

Q3

Advanced circuits • Coding • Project mentoring • Inter-school challenge.

Q4

Capstone projects • Quantum Day • Assessment • Certification.

Long-term

Regional Quantum & DeepTech learning centres, competitions and higher-education partnerships.

20. Partnership Invitation

Electropro India invites schools to participate in the 2026–27 Quantum & DeepTech Education Program and create an early, responsible pathway into emerging technology.

Positioning

Electropro — Quantum & DeepTech Education Partner

Tagline

From Bits to Qubits — Building Future Innovators.

Next step

Conduct a school-specific orientation and prepare a customized implementation plan.

Contact

Electropro India | Technology & Innovation Education

School Partnership Desk | 2026–27